

Prachit Gupta

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EDUCATION

University of Illinois Urbana Champaign

Master of Science in Mechanical Engineering; **GPA: 4.0/4.0**

Urbana, IL

Aug 2025 – May 2027 (Expected)

Indian Institute of Technology Bombay

Mumbai, India

B.Tech. in Mechanical Engineering; **GPA: 3.67/4.0**

Nov 2021 – May 2025

Minor Degree in Systems and Controls

July 2022 – May 2025

TECHNICAL SKILLS

Programming: C/C++, Python, Arduino, Matlab, Lua, URDF, Xacro

Software Tools: ROS, ROS2, Git, MATLAB, Simulink, Solidworks, Eagle, Gazebo, Arduino, QGC, Mission Planner, CoppeliaSim, X-Plane, Rviz, Mavros, Ardupilot, PX4 $\LaTeX$

Frameworks: Gymnasium, TIVA, Mavros, OpenCv, Scikit-learn, Numpy, Matplotlib, helperOc, CVXPY, CasADI

RELEVANT COURSEWORK

Math and Computing: Microprocessors and Automation, Linear and Nonlinear Systems, Random Processes, Games and Information, Numerical Analysis, Linear Algebra, Machine Learning, Operations Modeling

Hands-on Courses: Microprocessors and Automatic Control Lab, Mechanical Measurements, Design of Mechatronic Systems

Online Certifications: [Aerial Robotics \(UPenn\)](#), [Introduction to Internet of Things \(UCI\)](#)

WORK EXPERIENCE

Multi-Agent Control with Safety Guarantees [github](#)

IIT Bombay, India

Dept. of Systems and Control Engineering | Advisor: Prof. Mayank Barnwal

Nov 24 - May 25

- Developed decentralized neural networks in **PyTorch** to ensure **safe multi-agent** control across varying scales.
- Jointly trained control policies and **barrier functions**, enhancing safety and scalability in complex environments.
- Leveraged **Large Language Models** (LLMs) to intuitively and effectively optimize **Reinforcement Learning** (RL) reward functions for **Deep -Q-Network** agents in an Intelligent driving scenario in a **human-centric** way.

Safewalk Robotics [github](#)

Queen's University, Canada

Dept. of Electrical and Computer Engineering | Advisor: Prof. Matthew Pan

May 24- July 24

- Developed **ROS** controllers for detection based dynamic object following on **Quadruped Unitree GO1**.
- Crafted **human-following** algorithms in **ROS2** for **Turtlebot4** using **depthai-blazepose** for skeletal detection
- Implemented **SLAM** and **Follow the Gap** based object avoidance using **ROS2** and **NAV2** for indoor navigation.
- Designed a hybrid **SL-based RL policy** framework for person-following robots in outdoor environments, robust to occlusions, and detection inaccuracies by training **deep neural networks** on a custom expanded dataset.

Advanced Safety and Control Systems for Autonomous Vehicles [github](#)

IIT Bombay, India

Dept. of Systems and Control Engineering | Advisor: Prof. Arpita Sinha

July 2023 – March 2023

- Developed a **safety verification system** for autonomous cars using **forward reachability**, ensuring collision-free navigation and generating optimized emergency trajectories using **convex optimization** in **CVXPY**
- Computed **avoid sets** for AVs, realized as the viscosity solution of the **Hamilton-Jacobi PDE** for safe **robot-human interactions**, using the **helperOc** package for **dynamic programming** and optimization.
- Integrated reachability-based safety guarantees into a **Model Predictive Controller** for overtaking maneuvers of autonomous vehicles, modeled as a **zero-sum differential game** between the robot and a human lead vehicle.

Communication and Control of Autonomous Drones

IIT Bombay, India

Dept. of Aerospace Engineering | Advisor: Prof. Dhwanil shukla

Feb '22 - Dec '22

- Senior Avionics Engineer** responsible for conducting state-of-the-art research to achieve full aerial autonomy.
- Employed wind data for **barometric compensation**, ensuring stable altitude and preventing sudden drops.
- Customized the **Ardupilot's** autopilot codebase to develop a custom mode enabling immediate motor shutdown.
- Developed **Minimum Snap** trajectories with optimized time segments using **constrained quadratic programming**

Unmanned Aerial System Competition (UAS) 2023, UK 📄

Feb '23 - Jun '22

- Achieved outstanding results, winning 3 awards: **Best Design**; **Scrutineering Award** for meeting supportability goals; **Advancement Award** for highest score among newcomers at Buckminster, UK
- Designed a **7 kg drone** for relief operations, capable of lifting 150,000 of payload & a flight of **20+ min**.
- Leveraged **Lua Scripting**, **Pymavlink**, and **RPI GPIO** interfaces for safe and reliable payload drop actuation
- Simulated the complete mission in **Gazebo** environment using **ROS** with **ArduPilot** as the flight control stack.
- Tested additional **safety features** like datalink loss, GC Fail Safe, and geo-fencing of autopilot for high-speed

International Aerial Robotics Competition (IARC), Norway ▶

Aug '22 - Oct '22

- Recognized by the organizers with **Special Mention** for the concept of a system of mother and daughter drone.
- Manufactured and assembled a **15kg** payload drone, Audrey, sized 2m x 2m with **Xavier NX** onboard computer
- Engineered the entire circuitry and developed C code for an autonomous replacement and gripper assembly
- Designed and successfully fabricated a **custom PCB** board using **EAGLE** software to ensure compact circuitry

Estimation on Lie Groups 🔗

Jan '24 - Apr '24

- Derived spacecraft attitude estimators and observers using **Wahba**, **Quest**, **Triad** and **Kalman filter** algorithm.
- Authored a review paper on a Model Predictive Tracking Control for Invariant Systems on Matrix **Lie Groups**, utilizing **stable embedding** into Euclidean spaces combining manifold-based motion description with **MPC constraint handling**.

UAV Control 📄

Apr '23 - May '23

- Implemented and tuned **Control-Lyapunov** function-based Non-linear **Backstepping coupled Sliding Mode Controller** for better tracking and robust performance against disturbances and changing parameters
- Achieved stable tracking even with an additional **45%** uncertainty in rotary inertia and disturbances.
- Modelled lateral and longitudinal autopilots for a model of **Fixed-wing** aircraft in **Matlab** using **Simulink**.

Inverted Pendulum

Sep '24 - Nov '24

- Developed a framework to swing up and balance an inverted pendulum using **energy-based nonlinear** control.
- Designed a **state-feedback** control system to balance the pendulum at its vertical upward position using **LQR**.
- Implemented **LQR-based** swing-up strategy for **cart-pole** system, achieving stabilization from downward to upward position in **4 seconds**.

Soft Robotic Gripper 🔗

Sep '24 - Nov '24

- Developed a **soft robotic** compliant two-finger gripper mechanism for high **adaptability** and gentle gripping.
- Implemented a mechanism for adaptive handling of food items using **topological** and **shape optimization**.
- Designed a **3-DOF robotic arm** using servo motors and a solenoid actuator to demonstrate gripper's efficacy.

Taxi Route Optimization

Jan '24 - Apr '24

- Achieved **90%** cab-to-customer allocation of multi-depot problem by optimizing dispatch strategies.
- Enhanced the dispatch algorithm using a profit function that balanced number of cabs and customer waiting times
- Modelled the capacitated open **vehicle routing** with Pulp, using **MTZ constraints**, and **OpenStreetMap** for visualisation.

Bubble Trouble 🔗

Dec '24

- Developed a single-player bubble shooter game in **C++** with **event-handling**, headers, structures, **classes**, and Standard Template Libraries features such as vectors, arrays, and XEvent, utilizing SimpleCPP packages.
- Enhanced gaming experience with **interactive features** like scoring, timer, health bar and dynamically varying bubble sizes.

Object Detection and Tracking using YOLO

Jan '23

- Achieved a **10%** performance improvement in object detection model via image augmentation and preprocessing
- Trained **YOLOv4** on the COCO dataset to achieve **98% accuracy** in person, vehicle, and object detection

Autonomous Maze Solver

Mar '22 - Jun '22

- Ideated and Developed **CAD** assembly of a **wall climbing bot** with a rocker-bogie like mechanism and a servo gripper to autonomously navigate, locate, and pick colored balls in a given arena using **Solidworks**.
- Crafted python scripts for implementing **bug0**, **bug1** and obstacle avoidance algorithms using **ROS** and Gazebo.

MENTORSHIP AND TEACHING EXPERIENCE

Teaching Assistant — Makersspace, MS101 🌐

Jan 2025 – Present

Responsible for monitoring and evaluating lab sessions for 100+ first-year students on AutoCAD.

Mentored 20+ students in building an RC drone from scratch as part of the course project.

Technical Advisor Team Aerove 🌐

April 2023 – Present

UMIC aims to facilitate technical start-ups and foster an atmosphere of innovation.

Recruited and mentored new members in foundational robotics tools and algorithms

Led training sessions for 15 new recruits in ROS, OpenCV, and drone building.

Solve for society

Dec 2023 – Jan 2023

Led projects in Solve-For-Society by Amazon's Innovation Story Foundation.

Guided teams from underprivileged schools to create solutions for people with disabilities.

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Programming: C/C++, Python, Arduino, Matlab, Lua, URDF, Xacro

Software Tools: ROS, ROS2, Git, MATLAB, Simulink, Solidworks, Eagle, Gazebo, Arduino, QGC, Mission Planner, CoppeliaSim, X-Plane, Rviz, Mavros, Ardupilot, PX4, L^AT_EX

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AWARDS & ACHIEVEMENTS

- Awarded **fellowship** by **Canadian** Government to pursue research at Queens University, Kingston ('21)
- Qualified as a **KVPY** Fellow and finished among **top 3 %** of applicants from a pool of **97000+** candidates ('21)
- Secured All India Rank **917** out of **150,000+** candidates in the **JEE Advanced** Exam ('21)
- Secured All India Rank **388** in **Joint Entrance Exam Mains** among over **1.4 million** applicants ('21)

REFERENCES

Prof. Mayank Baranwal

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Prof. Matthew Pan

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